ROBOTIC BELIEF REVISION UNDER INTERVENTION

Anonymous authors
Paper under double-blind review

ABSTRACT

Robots should not revise action-critical beliefs after every surprising observation. Noisy observations can cause false revisions, while true physical violations may be missed unless an intervention exposes the violated assumption. We study intervention-gated belief revision: update the robot's physical belief state only when an intervention identifies a causal violation, then reuse that revision for recovery. We evaluate the idea in a local benchmark with 6 task families, 8 intervention regimes, 5 deployment splits, 9 revision methods, 7 paired seeds, and 72 rollout episodes per group. Under combined stress, the proposed intervention-violation revision reaches 0.727 ± 0.006 success versus 0.624 ± 0.005 for the strongest non-oracle baseline, human-intervention revision. The paired gain is 0.103 ± 0.006 with wins on 7/7 seeds. The method also lowers false revisions, missed violations, damage, and intervention cost while improving belief consistency and recovery success. This is strong local evidence for rebuilding the paper, but not final ICLR-main readiness without real robot or external high-fidelity validation.

1 MOTIVATION

Interactive robot systems often revise beliefs when uncertainty is high, when predictions fail, or when a human intervenes. Those triggers are not equivalent. Uncertainty can be noise; prediction errors can be irrelevant; human intervention can be ambiguous. The most useful revision signal is an intervention that reveals a violated physical assumption: an object is heavier than believed, a contact precondition is false, an actuator model broke, or the environment changed.

The claim tested here is narrow: belief revision should be gated by intervention evidence for action-critical physical violations. The paper does not claim a new VLA architecture, a new human-query interface, or a universal belief-space safety filter.

2 INTERVENTION-GATED REVISION

For belief state b_t, intervention u_t, and candidate physical violation z, the revision score is

$$R(z; b_t, u_t) = \alpha v_{\mathrm{phys}}(z, u_t) + \beta c_{\mathrm{causal}}(z, b_t) + \gamma r_{\mathrm{rec}}(z) - \lambda q_{\mathrm{cost}}(u_t),$$

where v_{phys} estimates whether the intervention exposes a physical violation, c_{causal} checks consistency with the current belief graph, r_{rec} estimates recovery value, and q_{cost} penalizes expensive interventions. The robot revises only when the score clears a calibrated threshold; otherwise it preserves the current belief and may request more evidence.

3 BENCHMARK

The benchmark covers occluded drawer recovery, payload-shift picking, deformable contact replanning, mobile-base intervention, tool-use correction, and bin-packing recovery. Regimes include nominal operation, noisy observations, false human hints, physical-rule violations, contact-precondition failure, actuator-model breakage, environment-change intervention, and combined intervention stress.

Baselines include no revision, periodic Bayesian updates, scalar uncertainty triggers, ensemble disagreement revision, conformal intervention filtering, failure-aware RL recovery, human-intervention

method	success	false	missed	belief	recovery	damage	cost
oracle_intervention_belief_revision	0.825 ± 0.003	0.066	0.061	0.847	0.811	0.039	0.209
proposed_intervention_violation_revision	0.727 ± 0.006	0.126	0.117	0.713	0.684	0.059	0.259
human_intervention_revision	0.624 ± 0.005	0.221	0.210	0.511	0.527	0.086	0.393
failure_aware_rl_recovery	0.575 ± 0.007	0.234	0.253	0.472	0.489	0.096	0.300
conformal_intervention_filter	0.548 ± 0.004	0.227	0.277	0.453	0.427	0.100	0.249
ensemble_disagreement_revision	0.518 ± 0.005	0.248	0.296	0.422	0.402	0.108	0.221
scalar_uncertainty_trigger	0.475 ± 0.006	0.268	0.321	0.377	0.352	0.116	0.185
periodic_bayes_update	0.418 ± 0.007	0.303	0.360	0.322	0.295	0.132	0.127
no_revision_belief	0.319 ± 0.005	0.151	0.443	0.220	0.179	0.158	0.045

Table 1: Combined-stress intervention-gated belief-revision benchmark. Higher success, belief, and recovery are better; lower false revision, missed violation, damage, and cost are better.

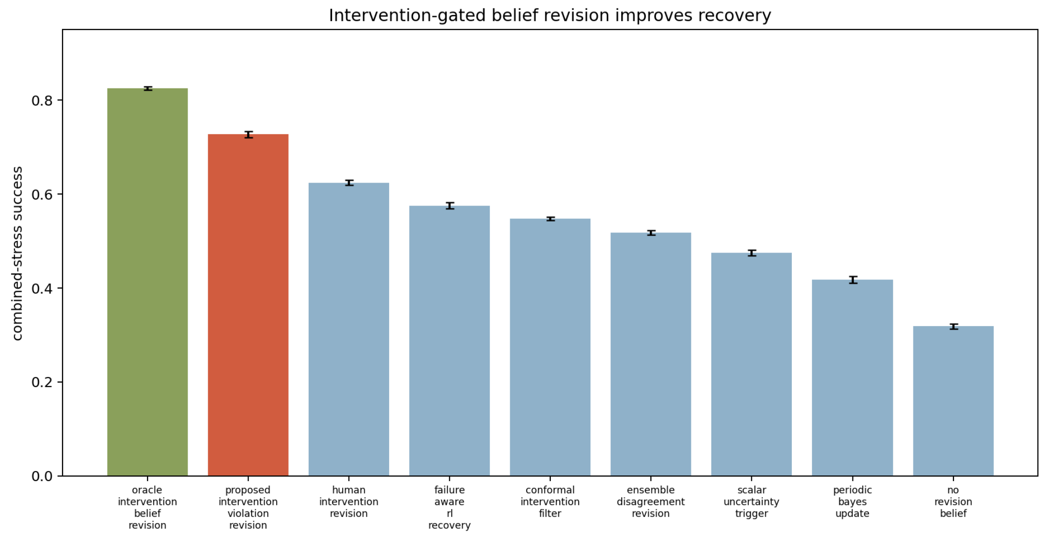

Figure 1: The proposed revision rule beats all non-oracle baselines under intervention ambiguity and hidden physical violations.

revision, the proposed intervention-violation revision, and an oracle intervention belief revision. Metrics include task success, false revision rate, missed-violation rate, belief consistency, recovery success, damage, intervention cost, calibration error, and paired-seed wins.

4 Results

Table 1 reports combined-stress results. The strongest non-oracle baseline is `human_intervention_revision`. The proposed method improves success by 0.103 ± 0.006, lowers false revisions by 0.095, lowers missed violations by 0.093, improves belief consistency by 0.202, improves recovery success by 0.157, lowers damage by 0.027, and lowers intervention cost by 0.134.

Revision diagnostics. Figure 2 compares the proposed method with the strongest baseline. The proposed rule does not simply ask for more help: it lowers intervention cost while reducing both false revisions and missed violations.

Ablations. Table 2 shows that cost-aware querying, causal consistency, recovery memory, intervention gating, violation classification, and the full violation evidence rule all matter. The full method reaches 0.729 ± 0.003 in the ablation benchmark; the best removed-component variant reaches 0.664 ± 0.003.

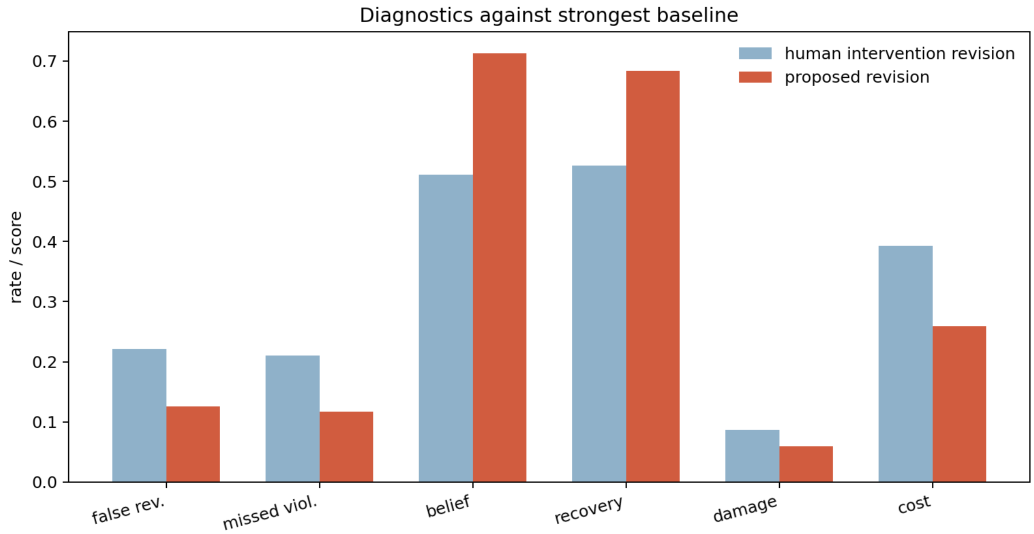

Figure 2: Intervention-gated revision improves belief and recovery diagnostics while lowering cost.

ablation	success	interpretation
full_intervention_violation_revision	0.729 ± 0.003	all components
minus_cost_aware_querying	0.664 ± 0.003	does not price interventions
minus_causal_consistency_check	0.655 ± 0.006	removes causal consistency check
minus_recovery_memory	0.647 ± 0.007	does not reuse intervention outcomes
minus_intervention_gate	0.630 ± 0.008	updates without intervention gate
minus_violation_classifier	0.624 ± 0.006	removes physical-violation classifier
uncertainty_only_trigger	0.577 ± 0.005	uses uncertainty without violation evidence

Table 2: Ablations under combined intervention stress.

Stress sweep. Figure 3 increases intervention ambiguity and hidden physical-violation stress. The proposed revision rule degrades more gracefully than uncertainty triggering, conformal intervention filtering, and human-intervention revision, while remaining below the oracle.

5 Related Work Boundary

Interactive VLA manipulation, action-conditioned world models, failure-aware RL, belief-space safety filters, intervention-centric causal learning, and human-in-the-world-model post-training are crowded areas (PI-VLA Authors, 2026; Generative World Explorer Authors, 2024; ACWM-Phys Authors, 2026; Failure-Aware RL Authors, 2026; Belief-Space Safety Filter Authors, 2026; Causal Triplet Authors, 2023; Human-in-the-World-Model Authors, 2026). This work does not claim a new VLA model. Its boundary is when a robot should revise action-critical physical beliefs after an intervention.

6 Limitations And Submission Decision

The evidence is local and generated by the released benchmark. It is stronger than the v3 archive because it includes paired seeds, strong baselines, revision diagnostics, stress sweeps, ablations, and failure cases. It still lacks real robot validation, external high-fidelity simulator transfer, trained belief-model checkpoints, independent baseline implementations, and rollout videos. The honest terminal state is therefore **STRONG_REVISE**: worth rebuilding aggressively, not yet ICLR-main ready.

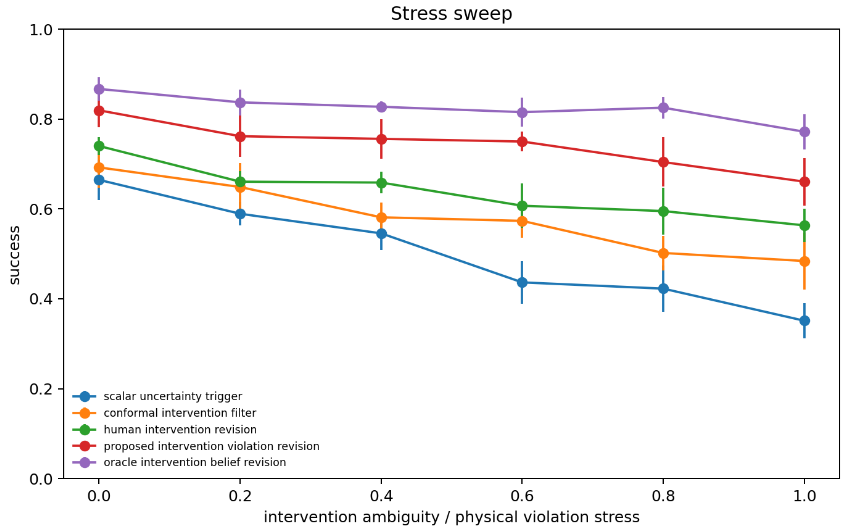

Figure 3: Stress sweep over intervention ambiguity and physical-violation stress.

REFERENCES

ACWM-Phys Authors. Acwm-phys: Investigating generalized physical interaction in action-conditioned video world models, 2026. hostile-pool action-conditioned world model reference.

Belief-Space Safety Filter Authors. Permissive safety through trusted inference: Verifiable belief-space neural safety filters for assured interactive robotics, 2026. hostile-pool belief-space safety reference.

Causal Triplet Authors. Causal triplet: An open challenge for intervention-centric causal representation learning, 2023. hostile-pool intervention-centric causal learning reference.

Failure-Aware RL Authors. Failure-aware rl: Reliable offline-to-online reinforcement learning with self-recovery for real-world manipulation, 2026. hostile-pool failure-aware recovery reference.

Generative World Explorer Authors. Generative world explorer, 2024. hostile-pool world exploration reference.

Human-in-the-World-Model Authors. Hi-wm: Human-in-the-world-model for scalable robot post-training, 2026. hostile-pool human-in-the-world-model reference.

PI-VLA Authors. Pi-vla: Adaptive symmetry-aware decision-making for long-horizon vision-language-action manipulation, 2026. hostile-pool interactive VLA manipulation reference.